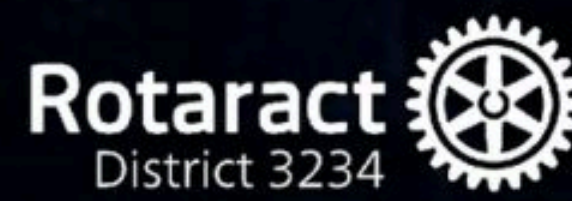




UNITE
FOR
GOOD



RISE
ABOVE



ROTA-TECH X

#INNOVATEFORIMPACT

TEAM DETAILS



URBAN PULSE

**AMRITA VISHWA
VIDYAPEETHAM**

DOMAIN: SUPPORTING THE ENVIRONMENT

PROBLEM STATEMENT:

Current civic monitoring systems depend on unverified complaints and manual processes, lacking validation, prioritization, and transparency. This results in misinformation, delayed issue resolution, and inefficient, unsustainable urban infrastructure management.

THE PROBLEM:

- Urban civic infrastructure monitoring is **inefficient and unreliable**
- Relies heavily on manual inspections and citizen-reported complaints
- Complaints are **often unverified, duplicated, or low in quality**
- **No proper system** to validate whether issues exist at the reported location

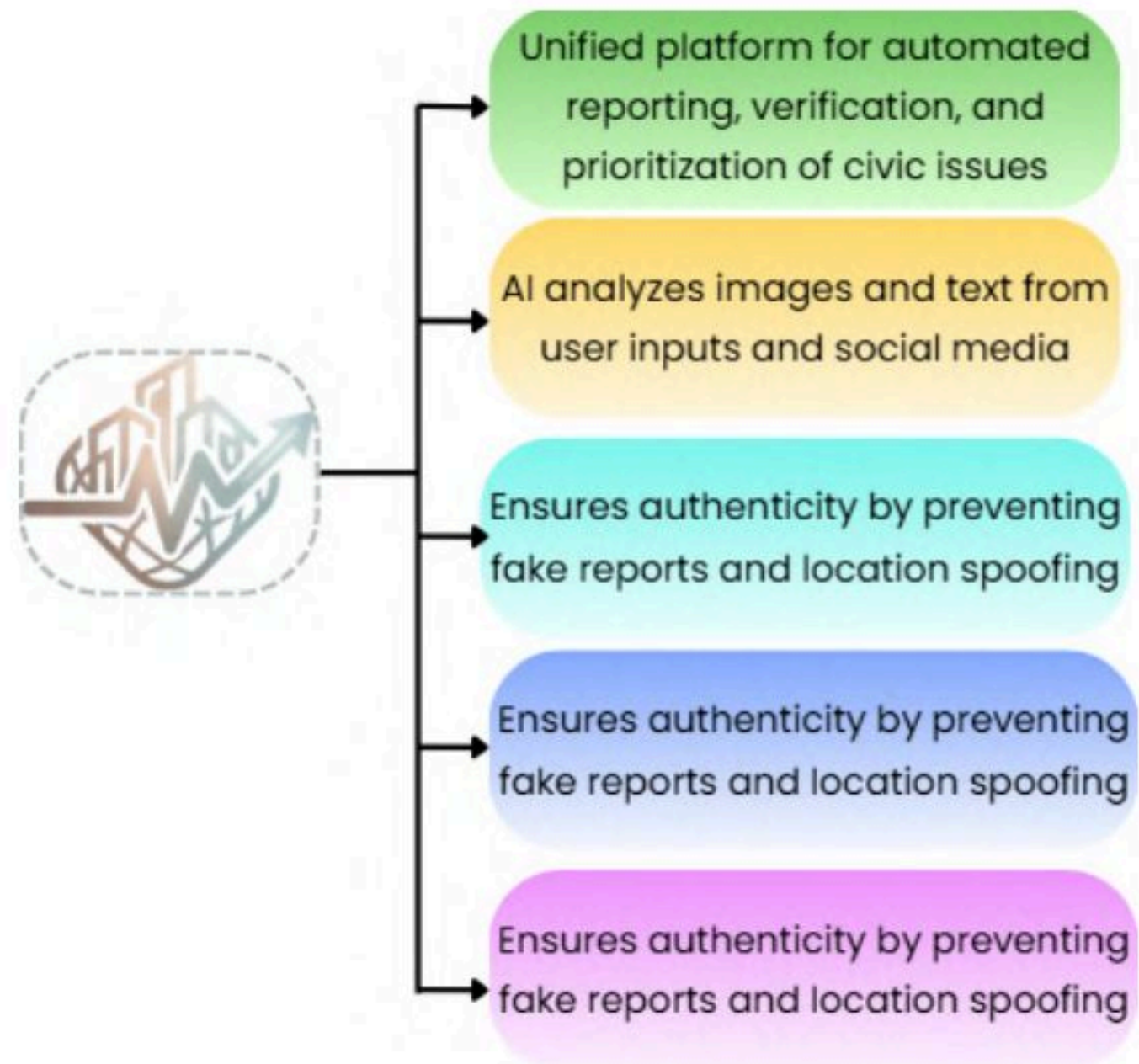
- **No structured method** to assess severity or prioritize issues
- Leads to **delayed responses and reactive maintenance**
- Results in higher long-term costs
- **Lack of transparency** reduces trust between citizens and authorities
- Vulnerable to **fake reports, reused images, and location spoofing**

HOW THE PROBLEM RELATES TO SUPPORTING THE ENVIRONMENT:

Poor monitoring and delayed resolution of civic issues lead to pollution, unsafe conditions, and inefficient resource use.

Addressing these gaps enables **cleaner, safer, and more sustainable urban environments** through data-driven management.

HOW URBANPULSE ADDRESSES THE PROBLEM



KEY FEATURES AND INNOVATIONS

Automated Monitoring Platform	End-to-end system for detection, verification, and prioritization
Multi-Source Data Collection	Inputs from citizen uploads and social media
NLP & Sentiment Analysis	Extracts issues, urgency, and trends from text data
AI-Based Issue Detection	Computer Vision for damage detection and image validation
Fraud & Authenticity Checks	Detects fake reports, location spoofing, and VPN misuse
Severity & Priority Analysis	Categorizes issues (Low / Medium / High) based on risk and urgency
Explainable AI (XAI)	Transparent reasoning behind decisions
Predictive Maintenance	Uses historical data to anticipate future issues

TECHNOLOGIES USED AND THEIR ROLE IN BUILDING THE SOLUTION:

AI & Machine Learning

- **Computer Vision (CNN Models)**
Detects infrastructure damage and validates images
- **NLP (DistilBERT)**
Extracts issues, sentiment, and urgency from text data
- **Explainable AI (Grad-CAM)**
Provides visual explanations for AI decisions

APIs & Integrations

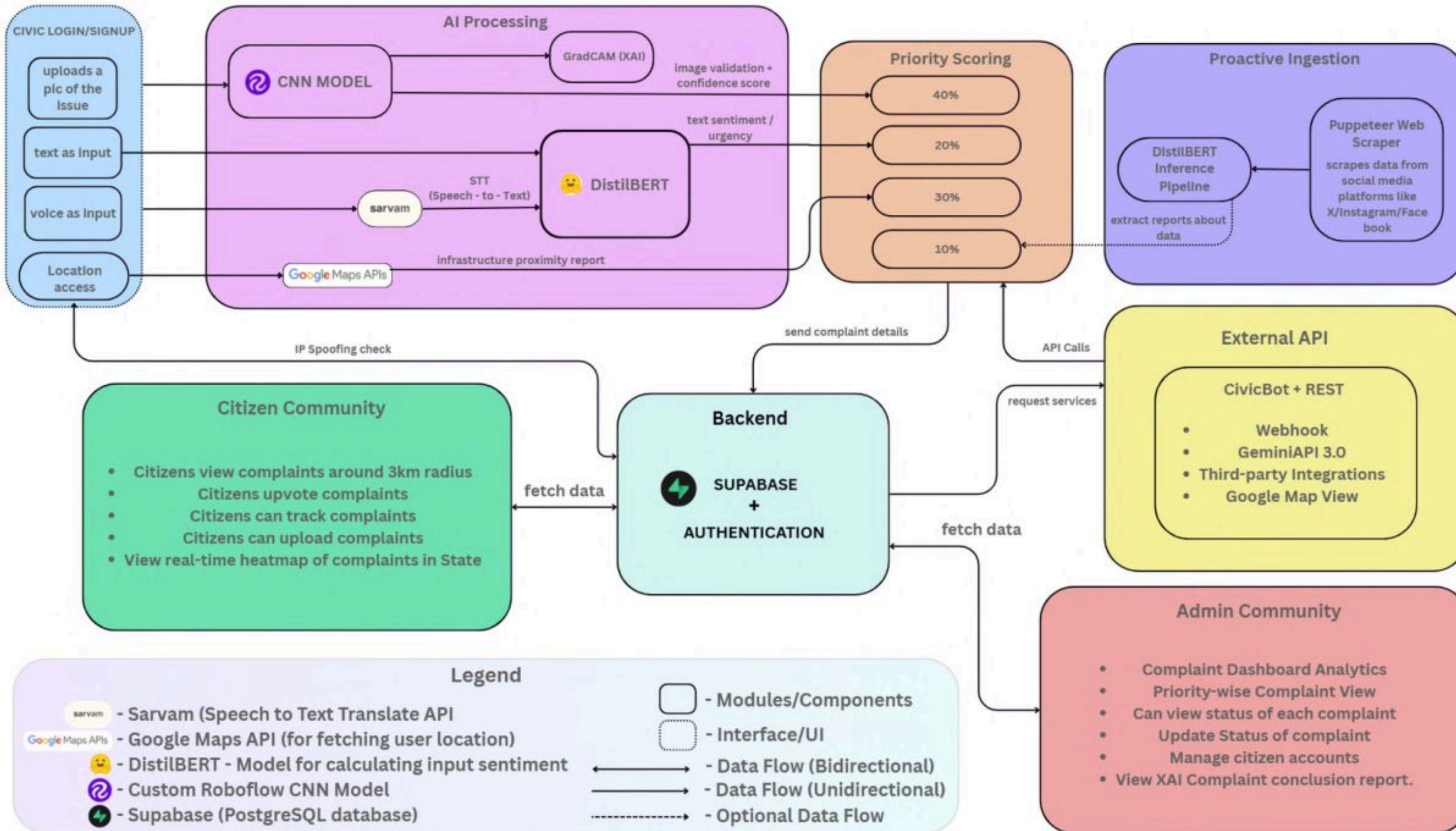
- **Google Maps API**
Location tracking and issue visualization
- **Sarvam Speech-to-Text API**
Converts voice inputs into text for accessibility
- **Puppeteer (Web Scraping)**
Extracts civic issues from social media platforms

Backend

- **Node.js / Express**
Handles APIs, data processing, and system logic
- **Supabase (PostgreSQL)**
Stores complaints, user data, and analytics

Frontend

- **React Native / Expo**
Mobile interface for submitting and tracking complaints



Social Impact:

- Improved Public Safety through early detection of hazardous infrastructure issues
- Faster Response Times via automated verification and prioritization
- Cost Savings for Municipalities by reducing manual inspections and emergency repairs
- Increased Transparency & Trust using explainable AI-based decision support
- Inclusive Urban Governance by capturing complaints from underserved areas and social media
- Sustainable Infrastructure Management through predictive maintenance

Feasibility:

- Easily deployable using existing infrastructure like mobile apps, GPS, and cloud platforms, requiring minimal additional setup
- Integrates smoothly with current municipal systems and workflows for real-time complaint handling and decision-making
- Scalable to handle large volumes of urban data and complaints across cities without performance issues
- Cost-effective solution by reducing manual inspections and leveraging open-source AI models and cloud services

NAME	ROLE	UNIVERSITY
ARUN N	Frontend & UI/UX Developer	AMRITA VISHWA VIDYAPEETHAM
KIRAN KISHORE VENKATESAN	AI & Machine Learning Engineer	AMRITA VISHWA VIDYAPEETHAM
VIMAL HARIHAR S K	Backend & Database Engineer	AMRITA VISHWA VIDYAPEETHAM
KESHAV S	Geo-Validation & System Integration Engineer	AMRITA VISHWA VIDYAPEETHAM
Yashwanth P T	Data Processing & Analytics Engineer	AMRITA VISHWA VIDYAPEETHAM